SIMATS ENGINEERING

Assessment Tool 3 — Mock Interview

4 Selected Answers (3-4 lines each)

Q1. How would you use Bayesian Networks to predict machine failures?

I would build a Bayesian Network linking sensor readings like temperature and vibration to a 'failure' node, with arrows showing cause and effect. Each link has a probability learned from past data. So if vibration rises, the network updates the failure chance instantly. This lets us do maintenance early, before a real breakdown.

Q3. How would you apply a Hidden Markov Model for speech recognition?

Speech is a sequence of sounds, so I'd use an HMM where hidden states are phonemes and observations are audio features. The model learns how sounds follow each other and how audio matches each sound. Using the Viterbi algorithm, I find the most likely sequence of sounds and match it to words.

Q9. Why do CRFs often outperform HMMs in named entity recognition?

HMMs only let each word depend on its own hidden label, ignoring other clues. CRFs look at the whole sentence and use overlapping features like capitalization and nearby words together. Since names often follow such patterns, this extra context helps CRFs tag names more accurately than HMMs.

Q5. How does conditional independence simplify probabilistic inference?

Conditional independence means once we know one variable, some others no longer affect our answer, so we don't need the full joint probability. For example, Grass Wet depends only on Rain and Sprinkler directly, not everything else. This lets us break big calculations into small local ones, using only nearby connections.